

Experiment 4

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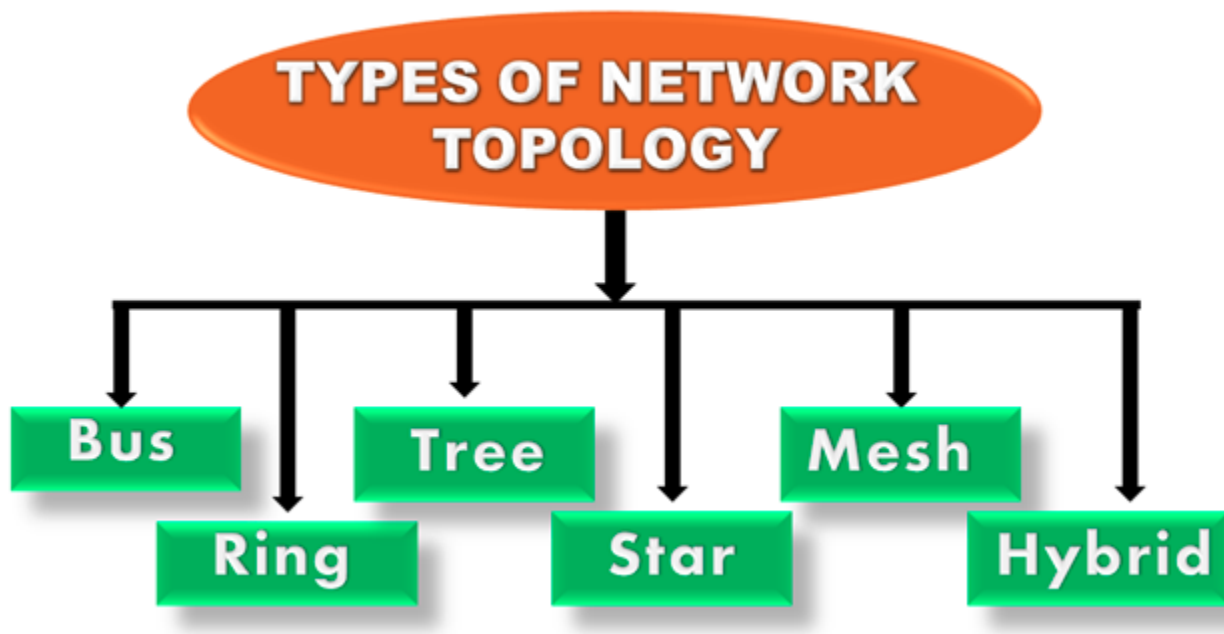
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Aim:-

Write TCL scripts to create topologies. Create and run traffics and analyze the result using NS2.

Theory:-

Topology defines the structure of the network of how all the components are interconnected to each other. There are two types of topology: physical and logical topology.



Bus:-

- The bus topology is designed in such a way that all the stations are connected through a single cable known as a backbone cable.
- Each node is either connected to the backbone cable by drop cable or directly connected to the backbone cable.
- When a node wants to send a message over the network, it puts a message over the network. All the stations available in the network will receive the message whether it has been addressed or not.

Ring:-

- Ring topology is like a bus topology, but with connected ends.
- The node that receives the message from the previous computer will retransmit to the next node.
- The data flows in one direction, i.e., it is unidirectional.
- The data flows in a single loop continuously known as an endless loop.
- It has no terminated ends, i.e., each node is connected to other node and having no termination point.
- The data in a ring topology flow in a clockwise direction.

Tree:-

- Tree topology combines the characteristics of bus topology and star topology.
- A tree topology is a type of structure in which all the computers are connected with each other in hierarchical fashion.
- The top-most node in tree topology is known as a root node, and all other nodes are the descendants of the root node.
- There is only one path exists between two nodes for the data transmission. Thus, it forms a parent-child hierarchy.

Star:-

- Star topology is an arrangement of the network in which every node is connected to the central hub, switch or a central computer.
- The central computer is known as a **server**, and the peripheral devices attached to the server are known as **clients**.

Mesh:-

- Mesh technology is an arrangement of the network in which computers are interconnected with each other through various redundant connections.
- There are multiple paths from one computer to another computer.
- It does not contain the switch, hub or any central computer which acts as a central point of communication.
- The Internet is an example of the mesh topology.

Hybrid:-

- The combination of various different topologies is known as **Hybrid topology**.
- A Hybrid topology is a connection between different links and nodes to transfer the data.
- When two or more different topologies are combined together is termed as Hybrid topology and if similar topologies are connected with each other will not result in Hybrid topology. For example, if there exist a ring topology in one branch of ICICI bank and bus topology in another branch of ICICI bank, connecting these two topologies will result in Hybrid topology.

Implementation:-

```
#create a new simulator object
set ns [new Simulator]
```

```
#open the nam trace file
set nf [open out.nam w]
$ns namtrace-all $nf
```

```
#define a finish procedure
proc finish {} {
    global ns nf
    $ns flush-trace
```

```
#close the trace file
close $nf
```

```
#execute nam on the trace file
exec nam out.nam &
```

```
exit 0
}
```

```
#create two nodes
set n0 [$ns node]
set n1 [$ns node]
set n2 [$ns node]
set n3 [$ns node]
set n4 [$ns node]
set n5 [$ns node]
set n6 [$ns node]
set n7 [$ns node]
set n8 [$ns node]
set n9 [$ns node]
set n10 [$ns node]
set n11 [$ns node]
set n12 [$ns node]
set n13 [$ns node]
set n14 [$ns node]
set n15 [$ns node]
set n16 [$ns node]
set n17 [$ns node]
set n18 [$ns node]
set n19 [$ns node]
set n20 [$ns node]
set n21 [$ns node]
```

```
#create a duplex link between the nodes
$ns duplex-link $n0 $n1 1Mb 10ms DropTail
$ns duplex-link $n8 $n1 1Mb 10ms DropTail
$ns duplex-link $n9 $n1 1Mb 10ms DropTail
$ns duplex-link $n10 $n1 1Mb 10ms DropTail
$ns duplex-link $n0 $n2 1Mb 10ms DropTail
$ns duplex-link $n2 $n3 1Mb 10ms DropTail
$ns duplex-link $n3 $n14 1Mb 10ms DropTail
$ns duplex-link $n3 $n11 1Mb 10ms DropTail
$ns duplex-link $n3 $n13 1Mb 10ms DropTail
$ns duplex-link $n3 $n12 1Mb 10ms DropTail
$ns duplex-link $n2 $n4 1Mb 10ms DropTail
$ns duplex-link $n4 $n5 1Mb 10ms DropTail
$ns duplex-link $n5 $n16 1Mb 10ms DropTail
$ns duplex-link $n5 $n15 1Mb 10ms DropTail
$ns duplex-link $n5 $n17 1Mb 10ms DropTail
$ns duplex-link $n4 $n6 1Mb 10ms DropTail
$ns duplex-link $n6 $n7 1Mb 10ms DropTail
$ns duplex-link $n20 $n7 1Mb 10ms DropTail
$ns duplex-link $n19 $n7 1Mb 10ms DropTail
```

\$ns duplex-link \$n18 \$n7 1Mb 10ms DropTail
\$ns duplex-link \$n21 \$n7 1Mb 10ms DropTail
\$ns duplex-link \$n6 \$n0 1Mb 10ms DropTail

\$ns duplex-link-op \$n14 \$n3 orient left-down
\$ns duplex-link-op \$n11 \$n3 orient right-down
\$ns duplex-link-op \$n13 \$n3 orient right
\$ns duplex-link-op \$n12 \$n3 orient right-up
\$ns duplex-link-op \$n3 \$n2 orient right
\$ns duplex-link-op \$n2 \$n0 orient right
\$ns duplex-link-op \$n2 \$n4 orient down
\$ns duplex-link-op \$n4 \$n6 orient right
\$ns duplex-link-op \$n0 \$n6 orient down
\$ns duplex-link-op \$n4 \$n5 orient right-down
\$ns duplex-link-op \$n16 \$n5 orient right
\$ns duplex-link-op \$n15 \$n5 orient right-up
\$ns duplex-link-op \$n17 \$n5 orient up
\$ns duplex-link-op \$n6 \$n7 orient right-down
\$ns duplex-link-op \$n7 \$n20 orient right-up
\$ns duplex-link-op \$n7 \$n19 orient right
\$ns duplex-link-op \$n7 \$n18 orient right-down
\$ns duplex-link-op \$n7 \$n21 orient left-down
\$ns duplex-link-op \$n7 \$n20 orient right-up
\$ns duplex-link-op \$n0 \$n1 orient up
\$ns duplex-link-op \$n1 \$n9 orient left-up
\$ns duplex-link-op \$n1 \$n10 orient right-up
\$ns duplex-link-op \$n1 \$n8 orient right

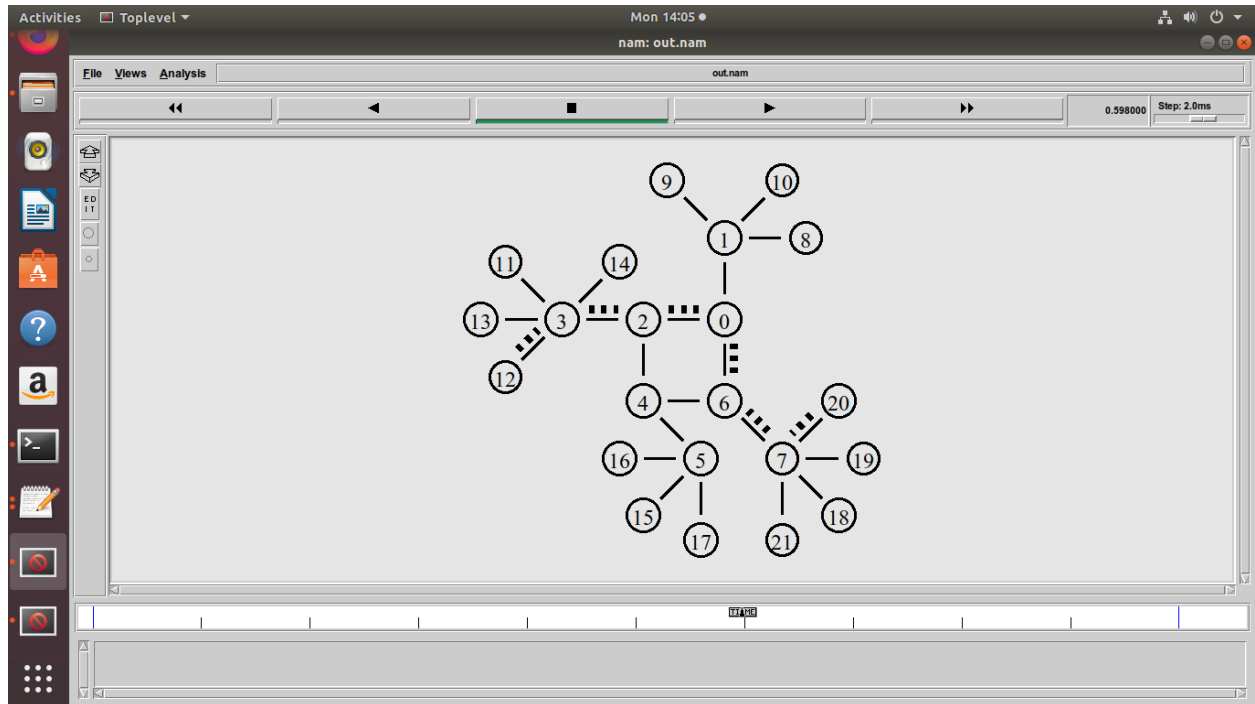
set udp [new Agent/UDP]
\$ns attach-agent \$n12 \$udp
set null [new Agent/Null]
\$ns attach-agent \$n20 \$null
\$ns connect \$udp \$null
\$udp set fid_ 2
set cbr [new Application/Traffic/CBR]
\$cbr attach-agent \$udp
\$cbr set type_ CBR
\$cbr set packet_ size 3000
\$cbr set random_ false

\$ns at 0.5 "\$cbr start"
\$ns at 3.0 "\$cbr stop"

\$ns at 1.0 "finish"

\$ns run

Output:-



Conclusion:-

TCL scripts used to create topologies and generate traffic.